

AsMA Panel
17-21 May, 2026
Denver, CO

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Panel #: 15177

BEHAVIORAL COMPONENTS OF NASA'S 2025 ASTRONAUT SELECTION CYCLE

BACKGROUND: Behavioral suitability is a critical determinant of astronaut performance and mission success, particularly for exploration missions involving isolation, confinement, operational complexity, and communication latency. As NASA advances the Artemis program and prepares for sustained lunar operations and future Mars exploration, the behavioral demands placed on crews continue to evolve. Aerospace Medical Association (AsMA) attendees—especially those working in aerospace medicine, operational psychology, and human performance—benefit from understanding how contemporary astronaut selection processes address these demands. This topic is timely following the recent completion of NASA's 2025 astronaut selection cycle, which emphasized behavioral competencies required for deep-space missions beyond low Earth orbit.

OVERVIEW: In 2025, NASA completed selection of its 24th class of astronaut candidates using a structured, multi-phase U.S. astronaut selection process designed to identify individuals best suited for the medical, psychological, and behavioral demands of long-duration spaceflight. More than 8,000 applications were received during this cycle. An initial subset of highly qualified candidates was invited to NASA Johnson Space Center (JSC) for preliminary screening, which included a first Astronaut Selection Board (ASB) interview, standardized medical examinations, and psychological testing. A smaller group advanced to a second and final phase involving comprehensive medical evaluations, in-depth behavioral assessments, psychiatric examinations, and a second ASB interview. From this process, 10 astronaut candidates (ASCANs) were selected to begin basic training in Houston in September 2025. This presentation will review the structure, content, and implementation of the behavioral components embedded within this phased selection process, as well as key lessons learned.

DISCUSSION: The multi-phase behavioral selection framework enhances operational reliability by systematically integrating principles from operational psychology, aerospace psychology, and aerospace psychiatry within a high-stakes selection environment. When conducted by clinicians with direct spaceflight operations experience, this approach supports early identification of behavioral strengths,

potential vulnerabilities, and team compatibility relevant to exploration mission environments. The process advances aerospace medicine and human performance by reducing behavioral and psychiatric risk across mission phases and by aligning selection criteria with evolving exploration-class mission requirements. Lessons learned from NASA's 2025 astronaut selection cycle are applicable beyond the civilian space program and may inform behavioral selection practices across military, international, and commercial spaceflight organizations, supporting greater standardization and cross-domain collaboration within the aerospace community.

LEARNING OBJECTIVES:

1. Describe the structure and rationale of the multi-phase behavioral evaluation process used during NASA's 2025 astronaut selection cycle.
2. Identify behavioral and psychological factors relevant to astronaut suitability for long-duration and deep-space missions.

CME QUESTIONS (with Answers):

1. **True or False:** NASA's 2025 astronaut selection process included multiple behavioral assessment phases integrated with medical and ASB evaluations.

Answer: True

2. Which factor is *most* relevant to behavioral suitability for long-duration deep-space missions?
 - A. Technical specialization alone
 - B. Team compatibility and stress tolerance
 - C. Age at time of selection
 - D. Prior spaceflight experience only

Answer: B

3. The behavioral selection framework used in NASA's 2025 cycle is most directly intended to:
 - A. Eliminate the need for in-flight behavioral support
 - B. Reduce psychological and operational risk across mission phases
 - C. Replace medical screening criteria
 - D. Standardize international astronaut training curricula

Answer: B